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Date of showing evaluated answer books:

No. of Printed Pages: 4

Roll No. ....

HARCOURT BUTLER TECHNICAL UNIVERSITY, KANPUR

M.C.A.

End Semester Examination

Even Semester (II), 2021-22

ECA-456: Database Management System

Time: 2:30 Hours **Max. Marks: 50**

Note: 1. Attempt all questions. All questions carry marks, as shown against them.

Course Outcomes (CO) in statement form

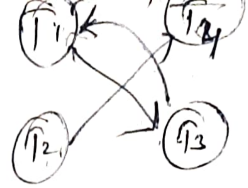
1. Understand and Develop Entity Relationship (ER) and Relational Models for a given application.
2. Develop and manipulate relational database using Structured Query Language and relational languages.
3. Develop a normalized database for a given application by incorporating various constraints like integrity and value constraints.
4. Understand and apply transaction processing concepts and convert schedules to serializable schedules.
5. Illustrate different concurrency control mechanisms to preserve data consistency in a multi-user environment.

		Related Course Outcome (CO)	Marks
Q. No. 1:	Attempt both questions.		
(a)	Compare DBMS with early file systems bringing out the major advantage of DBMS approach.	CO1	**(04)**
(b)	Draw an E-R diagram for a bank database with a set of customer and a set branch and account, with each customer a log of the various accounts also associated.	CO1	**(04)**

OR

Define the following with an example:
i) Weak entity type
ii) Specialization
iii) Key
iv) Multi valued attribute

Q. No.2:	Attempt both questions.		
(a)	Discuss the selection, projection and join operator of relational algebra with suitable examples.	CO2	**(04)**
(b)	Explain the primary key, Super key, Foreign key and Candidate key with example.	CO2	**(04)**
Q. No. 3:	Attempt both questions.		
(a)	Explain 1NF, 2NF with examples.	CO3	**(04)**
(b)	Write the difference between 3NF and BCNF. Find normal form of relation R(A, B, C, D) having Functional dependency set F= {A → B, BC → E, ED → A}	CO3	**(04)**



Q. No. 4: Attempt both questions.

(a) What do you mean by schedule in the context of concurrent execution of transaction in RDBMS? Discuss the various types of serializability with a suitable example. CO4 **(04)**

(b) What is deadlock? What are necessary conditions for it? How it can be detected and recovered? CO4 **(04)**

Q. No. 5: Attempt both questions.

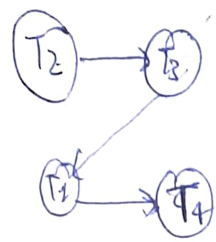
(a) What do you mean by time stamping protocols for concurrency control? Discuss multi-version scheme of concurrency control also. CO5 **(04)**

(b) Check whether the given schedule is serializable and recoverable or not. CO4 **(04)**

T1	T2	T3	T4
	R(X)		
		W(X)	
		Commit	
W(X)			
Commit			
	W(Y)		
	R(Z)		
	Commit		
			R(X)
			R(Y)
			Commit

$T_2 \to T_4$

recoverable

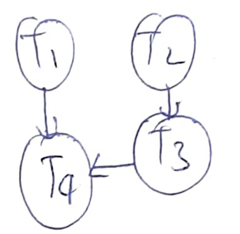


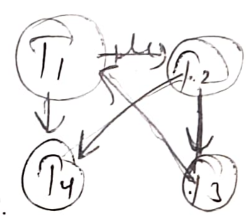
$T_2\ T_3\ T_1\ T_4$

Q. No. 6: Attempt all questions.

(a) CO2 **(05)**

Empid	EmpName	Department	ContactNo	EmailId	EmpHeadId
101	Isha	E-101	1234567890	isha@gmail.com	105
102	Priya	E-104	1234567890	priya@yahoo.com	103
103	Neha	E-101	1234567890	neha@gmail.com	101
104	Rahul	E-102	1234567890	rahul@yahoo.com	105
105	Abhishek	E-101	1234567890	abhishek@gmail.com	102





A) Write SQL query to create the above table.

B) Write SQL query to Select the detail of the employee whose name start with P.

C) Write a SQL query to select the detail of employee whose emailId is in gmail.

D) Select the name of the employee whose name's 3rd character is 'h'.

E) Write a SQL query to fetch the names of candidates who withdraw salary more than 5000.

(b) Write SQL query for the given table. CO2 **(05)**

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
001	Monika	Arora	100000	2014-02-20 09:00:00	HR
002	Niharika	Verma	80000	2014-06-11 09:00:00	Admin
003	Vishal	Singhal	300000	2014-02-20 09:00:00	HR
004	Amitabh	Singh	500000	2014-02-20 09:00:00	Admin
005	Vivek	Bhati	500000	2014-06-11 09:00:00	Admin
006	Vipul	Diwan	200000	2014-06-11 09:00:00	Account
007	Satish	Kumar	75000	2014-01-20 09:00:00	Account
008	Geetika	Chauhan	90000	2014-04-11 09:00:00	Admin

A) Write an SQL query to fetch "FIRST_NAME" from Worker table using the alias name as <WORKER_NAME>.
B) Write an SQL query to fetch "FIRST_NAME" from Worker table in upper case.
C) Write an SQL query to fetch unique values of DEPARTMENT from Worker table.
D) Write an SQL query to print the first three characters of FIRST_NAME from Worker table.
E) Write an SQL query to find the position of the alphabet ('a') in the first name column 'Amitabh' from Worker table.

Harcourt Butler Technical University Kanpur			END SEM EXAM
A	M.C.A.	Program	M.C.A.
rse Name	Database Management System	Semester	II
ourse Code	ECA-456	Year	2022-2023
Time:	02:30 Hr	Maximum Marks	50

Knowledge Level (KL)			
Knowledge Level (KL)	**K1**:Remembering	**K3**:Applying	**K5**:Evaluating
	K2:Understanding	**K4**:Analysing	**K6**:Creating

Note: Answer All Questions

Q.No	Questions	Marks	COs	KL
1	Attempt **both** questions.			
(a)	What is Database? Define characteristics of DBMS. How is DBMS different from File system?	4	CO1	K1
(b)	What are different types of anomalies associated with database? In a given relation R(A1A2…………An), how many super keys are possible if : • A1 is candidate key. • A1, A2 are candidate keys. **OR** Construct an E-R diagram for a car-insurance company whose customer own one or more cars each. Each car has associated with it zero to any number of recorded accidents.	4	CO1	K6
2.	Attempt **both** questions.			
(a)	Define the following with suitable example: i. Natural Join ii. Equi join	4	CO2	K1, K2
(b)	Consider the following relational database schema consisting of the four relation schemas: Passenger(pid, pname, pgender, pcity) Agency(aid, aname, acity) Flight(fid, fdate, time, src, dest) Booking(pid, aid, fid, fdate) Answer the following using relational algebra queries: i. Get the complete details of all flights to New Delhi. ii. Get the details about all flights from Chennai to New Delhi. iii. Find the passenger names for passengers who have bookings on at least one flight. iv. Find the agency names for agencies that located in the same city as passenger with passenger id 123.	4	CO2	K3
3.	Attempt **both** questions:			
(a)	Why do we normalise database? Explain BCNF and 2NF with examples.	4	CO3	K1, K2
(b)	Given a relation R(P,Q,R,S,T,U,V,W,X,Y) and FD = {PQ → R, P → ST, Q → U, U → VW, S → XY}, determinbe whether the given R is in 3NF? If not then convert it into 3NF.	4	CO3	K4, K5, K6

<table>
<tr><td>4.</td><td colspan="2">Attempt both questions.</td><td></td><td></td><td></td></tr>
<tr><td></td><td>(a)</td><td>What do you understand by transaction? What are operations performed in database during transaction? Define the ACID properties with suitable example.</td><td>4</td><td>CO</td><td></td></tr>
<tr><td></td><td>(b)</td><td>What do you mean by serializability? Check whether the given schedule id conflict serializable or not. If not then convert it into view serializable.
T1 T2T3
<table>
<tr><th>T1</th><th>T2</th><th>T3</th></tr>
<tr><td>R(A)</td><td></td><td></td></tr>
<tr><td></td><td>W(A)</td><td></td></tr>
<tr><td>W(A)</td><td></td><td></td></tr>
<tr><td></td><td></td><td>W(A)</td></tr>
</table></td><td>4</td><td>CO4</td><td>K4</td></tr>
<tr><td>5.</td><td colspan="2">Attempt both questions:</td><td></td><td></td><td></td></tr>
<tr><td></td><td>(a)</td><td>What is need of lock in DBMS? Explain shared lock and exclusive lock with the help of example.</td><td>4</td><td>CO5</td><td>K2</td></tr>
<tr><td></td><td>(b)</td><td>What do you understand by Time stamping? Define itwith suitable example.</td><td>4</td><td>CO5</td><td>K3</td></tr>
<tr><td>6.</td><td colspan="2">Attempt both parts:</td><td></td><td></td><td></td></tr>
<tr><td></td><td>(a)</td><td>Consider the following relations:
Employee (<u>Eno</u>, name, dob, date_of_joining, designation, gender, basic_pay, dno)
Department (<u>dno</u>, dname)
The primary key is underlined. Write SQL queries to perform the following:
i. Display the employee number, name, department number and department name of all employees.
ii. List the details of employees who earn less than average basic pay of all employees.
iii. List the department number and number of employees in each department.</td><td>1
2
2</td><td>CO2</td><td>K5, K6</td></tr>
<tr><td></td><td>(b)</td><td>Consider the following relation. The Primary key is Rollno, Isbn,
Student(RollNo, Name, Branch),
Book(Isbn, Title, Author, Publisher)
Issue(Rollno, Isbn, te_of_issue). Write the SQL query for the following:
i) List the Roll Number and Name of All CSE Branch Student.
ii) Find the name of students who have issued a book of publication 'BPB'.
iii) List the title and author of all books which are issued by a student name started with a.
iv) List the title of all books issued on or before 20/09/2012.
v) List the name of student who will read the book of author named 'Sanjeev'.</td><td>5</td><td>CO2</td><td>K6</td></tr>
</table>

Course Outcomes	CO1	Understand and Develop Entity Relationship (ER) and Relational Models for a given application.
	CO2	Develop and manipulate relational database using Structured Query Language and relational languages.
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	CO4	Understand and apply transaction processing concepts and convert schedules to serializable schedules.
	CO5	Illustrate different concurrency control mechanisms to preserve data consistency in a multi-user environment.

multi-user environment.

CO4

Harcourt Butler Technical University Kanpur			END SEM
Branch	MCA	Program	MCA
Course Name	Database Management System	Semester	II
CourseCode	ECA-456	Year	2023-24
Time:	02:30 Hr	Maximum Marks	50

Knowledge Level(KL)	K1:Remembering	K3:Applying	K5:Evaluating
	K2:Understanding	K4:Analysing	K6:Creating

Note: All questions are compulsory.

Q.No	Questions	Marks	COs	KL
1	a) Explain ACID properties of Transaction.	4	CO4	K1
	b) What is an ER Diagram? Also, make an ER Diagram for Employee Project Management System.	4	CO1	K2, K6
2.	a) Illustrate the types of integrity constraints with the help of suitable examples.	5	CO2	K4
	b) Discuss the procedure of deadlock detection and recovery in transaction?	3	CO4	K1
3.	a) Find the highest normal form of a relation R(A, B, C, D) with a functional dependency set {B->A, A->C, BC->D, AC->BE} .	4	CO3	K3
	b) Describe the term MVD in the context of DBMS by giving an example.	4	CO1	K1
4.	a) Consider the following schedule for transaction **T1, T2, T3** and **T4**. S : $R_1(A)$, $W_1(B)$, $R_2(B)$, $R_3(C)$, $W_1(A)$, $R_4(A)$, $R_2(C)$, $W_4(A)$, $W_3(B)$, $R_4(B)$, $W_4(C)$, Check whether the given schedule S is conflict serializable or not.	5	CO4	K3
	b) What are different types of anomalies associated with database?	3	CO3	K2
5.	a) Explain Time Stamp Based Concurrency Control technique.	4	CO5	K2
	b) Discuss 2 phase commit (2PC) protocol and how the validation based protocols differ from 2PC?	4	CO4	K1, K2
6.	a) Write SQL queries for extracting data from more than one table (Equi-join, Non-Equi-join, Outer join).	5	CO2	K6
	b) Write SQL queries for Data Definition and Data Manipulation language.	5	CO2	K6

Course Outcomes	CO1	Understand and Develop Entity Relationship (ER) and Relational Models for a given application. (Understand, Apply)
	CO2	Develop and manipulate relational database using Structured Query Language and relational languages. (Apply)
	CO3	Develop a normalized database for a given application by incorporating various constraints like integrity and value constraints. (Apply)
	CO4	Understand and apply transaction processing concepts and convert schedules to serializable schedules. (Understand, Apply)
	CO5	Illustrate different concurrency control mechanisms to preserve data consistency in a multiuser environment. (Apply)

	Harcourt Butler Technical University Kanpur	END SEM EXAM

Branch	M.C.A.	Program	M.C.A.
Course Name	Database Management System	Semester	II
CourseCode	ECA-456	Year	I (2025)
Time:	02:30 Hr	MaximumMarks	50

Knowledge Level(KL)	K1:Remembering	K3:Applying	K5:Evaluating
	K2:Understanding	K4:Analysing	K6:Creating

Note: Answer All Questions

Captain

Q.No	Questions	Marks	COs	KL
1	Attempt **all** questions.			
(a)	A database is being constructed to keep track of the teams and games of a sport league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of players participating in each game for each team, the positions they play in that game and the result of the game.Design an E-R schema diagram for this application.	4	CO1	K1,K6
(b)	How is referencing table different from referenced table?	2	CO1	K1
(c)	Define the following and state that what will be the primary key in the relationship table of the following: i. One-to-many relationship ii. One-to-one relationship	2	CO1	K1, K2
2.	Attempt **both** questions.			
(a)	*Define the following with suitable example:* *i. Natural Join* *ii. Equi join*	4	CO2	K1,K2
(b)	*Consider the following relational database of a college:* **Student** (RollNo, Sname, add) **Teachers** (tid, tname, teaching subject) **College** (roll no, tid) Write relational algebra expression for the following: i. Find the name of the student who lives in Lalitpur. ii. Find the name of the teacher who teaches DBMS subject. iii. Find the name of the teacher who teaches CO subject to Rahul student. iv. Delete records of students whose address is "Lucknow".	4	CO2	K3,K4
3.	Attempt **both** questions:			
(a)	What do you understand by normalization? How is lossy decomposition different from lossless decomposition. Explain with an example.	4	CO3	K1
(b)	Given a relation R(A,B,C,D,E,F) and FD = {AB $\rightarrow$C, C $\rightarrow$ DE, E$\rightarrow$F, F$\rightarrow$A}. Check the highest normal form.	4	CO3	K4,K6
4.	Attempt **both** questions.			
(a)	What is Conflict Serializable Schedule? Check the given Schedule S1 is Conflict Serializable or not? S1: R4(A), R2(A), R3(A), W1(B), W2(A), R3(B), W2(B)	4	CO4	K4,K5
(b)	Define the following with suitable example: i. Strict recoverable schedule ii. Log based recovery	4	CO4	K1,K2

team ✓ game & sport league

position, partici, game, result, track of players

OR

(b)	Discuss the procedure of deadlock detection and recovery in transaction.			
5.	Attempt **both** questions:			
(a)	Discuss the timestamp ordering protocol for concurrency control. How does strict timestamp ordering differ from basic timestamp ordering?	4	CO5	K1
(b)	Discuss 2 phase commit (2PC) protocol and time stamp-based protocol with suitable example.	4	CO5	K1
6.	Attempt **both** parts:			
(a)	Consider the following schema: employee (ename, street, city) works (ename, comp_name, salary) company (comp_name, city) manages (ename, manager_name) Write the following queries in SQL: (i) Find the name of all employees in database who do not work for 'First Bank corporation'. Assume that all people work for one company. (ii) Find the names of all employees who earn more than every employee of 'Small Bank corporation'. (iii) Find the names, street address and cities of residence for all employees who work for 'First Bank corporation' and earn more than 50,000.	(2+1+2)	CO2	K6
(b)	Consider the following schema for institute library: Student (RollNo, Name, Father_ Name, Branch) Book (ISBN, Title, Author, Publisher) Issue (RollNo, ISBN, Date-of –Issue) Write the following queries in SQL: (i) List roll number and name of all students of the branch 'CSE'. (ii) Find the name of student who has issued a book published by 'ABC' publisher. (iii) List title of all books and their authors issued to a student 'RAM'. (iv) List title of all books issued on or before December 1, 2020. (v) List all books published by publisher 'ABC'	5	CO2	K6

Coordinate
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